

Aditya Menon

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EDUCATION

University of Waterloo

Bachelor of Mathematics in Computer Science, Digital Hardware Specialization

Sept. 2025 – Present

GPA 89/100

- **Relevant Coursework** : Advanced Data Structures and Algorithms, Advanced Linear Algebra, Advanced Calculus, OOP, Software Design and Testing, Digital Circuits and Systems
- **Scholarships** : Faculty of Mathematics Global Scholarship (\$18k CAD) | President's Scholarship of Distinction (\$2k CAD)

TECHNICAL SKILLS

Languages : C, C++, Python, TypeScript, JavaScript, SQL, Bash, Racket, Verilog

Frameworks & Tools : Next.js, React, Tailwind CSS, FastAPI, PostgreSQL, Docker, Git, GitHub Actions, OpenCV, Three.js, Linux, $\text{\LaTeX}$

Concepts : REST APIs, CI/CD, Embedded Systems, I²C, Real-Time Systems, Low-Level Debugging, Automated Testing

EXPERIENCE

UW Orbital

Onboarding, Firmware team

Apr. 2026 – Present

- Developing **embedded C firmware** for a satellite thermal monitoring subsystem, implementing an **I²C driver** for the LM75BD temperature sensor and low-level hardware interfaces for reliable telemetry collection.
- Designing periodic thermal-management logic and **interrupt-driven overtemperature handling**, integrating sensor polling, event processing, testing, and OS-level interrupt services in a constrained embedded environment.

PROJECTS

SecurePaste

- Built a full-stack secure secret-sharing web application using Next.js, TypeScript, FastAPI, and PostgreSQL, supporting encrypted storage, single-use access links, TTL-based expiry, and audit logging.
- Implemented REST APIs, secure token hashing, Dockerized local development, and pytest-based backend tests for one-time-read and expiry enforcement logic.
- Configured GitHub Actions CI pipelines for automated frontend and backend testing, validation, and streamlined development workflows.

CubeCoach

- Building a real-time computer vision system using OpenCV to detect Rubik's Cube states from webcam video and reconstruct puzzle state throughout a solve.
- Developing a coaching platform with Next.js and Three.js that analyzes solve sequences, tracks performance metrics, and delivers interactive post-solve feedback visualizations.

C-based Programming Language Interpreters

- Developed multiple programming language interpreters in C, including a memory-managed imperative-language interpreter and a substitution-based evaluator for a functional language.
- Designed and implemented a continuation-passing interpreter featuring custom continuation data structures and trampolining to eliminate recursive control flow, enabling efficient evaluation of deeply nested functional programs without stack growth.

Awards

Bronze Medallist, Indian National Olympiad in Informatics

Apr. 2025

- Bronze medallist at a national-level competitive programming Olympiad for high school students, held to select a team to represent the country at the International Olympiad of Informatics (IOI).

1st Place, International Collegiate Programming Contest, Qatar (QCPIC)

Sept. 2024

- Won 1st place in a national ICPC-style programming contest with 110+ teams, receiving a 2.25k CAD cash prize and recognition in the Gulf Times.

Top 25, ICPC local contest, University of Waterloo

Oct. 2025

- Placed 22nd among 100+ participants in the University of Waterloo ICPC local contest, held to select teams for Waterloo's internationally competitive ICPC program.

ACTIVITIES

TEDxYouth@DPSMIS 2024

Sept. 2024 – Dec. 2024

Head of IT

- Led technical operations for a large-scale TEDxYouth event, coordinating audio, lighting, presentation, and networking systems to ensure reliable execution of live sessions for speakers and attendees.
- Managed backstage technical infrastructure throughout the event, rapidly diagnosing and resolving unexpected hardware and software issues under time constraints while maintaining uninterrupted event operations.

High School Computer Club

Apr. 2024 – Jan. 2025

Director

- Led educational initiatives for the school's Computer Club, developing and presenting workshops on cybersecurity fundamentals and C++ programming to students with diverse technical backgrounds.
- Organized a beginner-friendly Capture the Flag cybersecurity competition from conception to execution, creating learning resources, conducting preparatory workshops, and managing event operations.